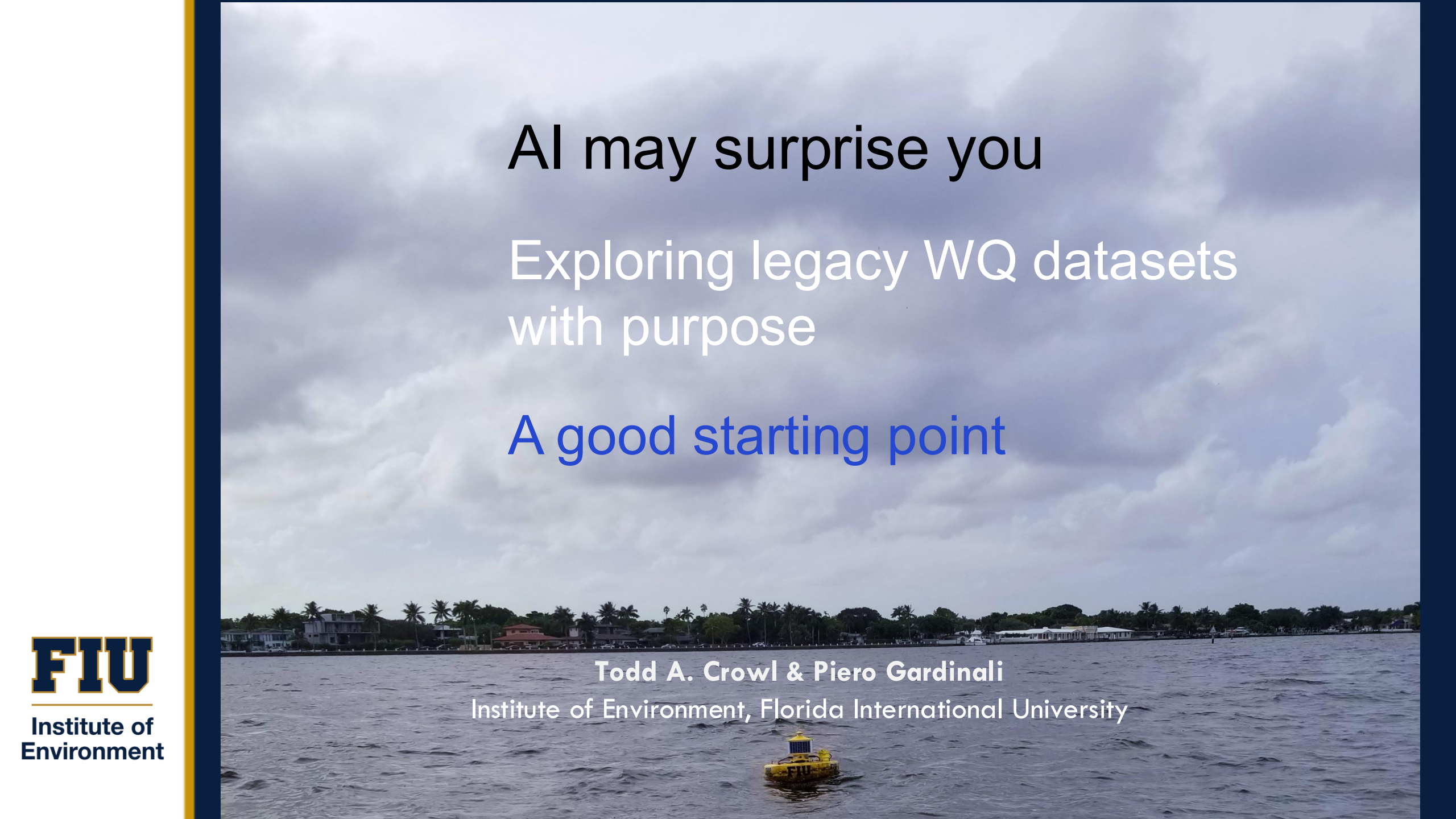




AI may surprise you

Exploring legacy WQ datasets with purpose

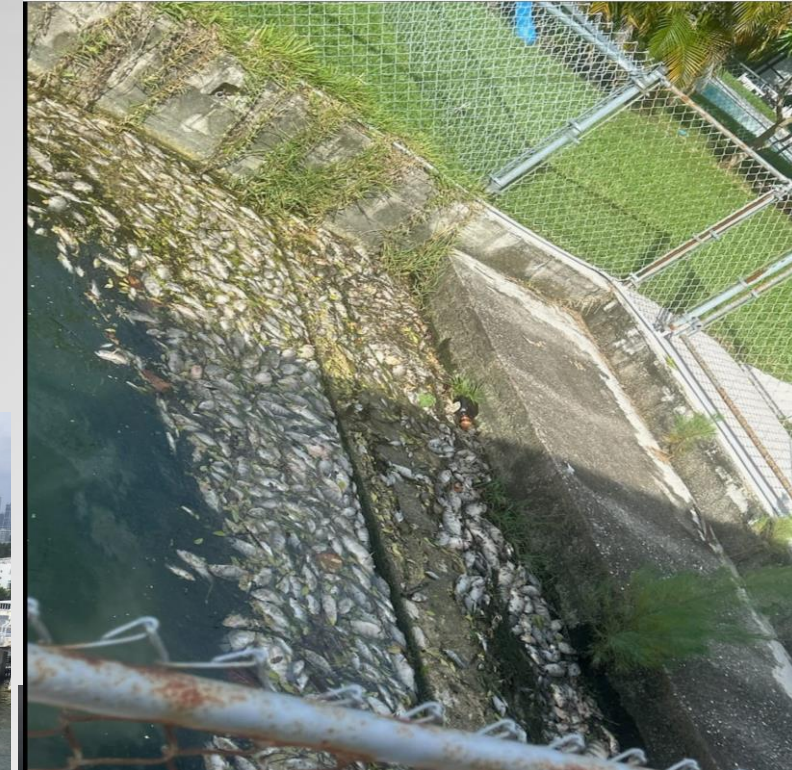
A good starting point

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Todd A. Crawl & Piero Gardinali
Institute of Environment, Florida International University







We had continuous monitoring with real-time data through our NSF Center of Excellence Program



Including night-time observations

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and Environment
NSF Center of Research Excellence
in Science and Technology

Environment.fiu.edu



Chart View

Table View

Site Information

Parameters ▾

Studies ▾

📅 Last Day

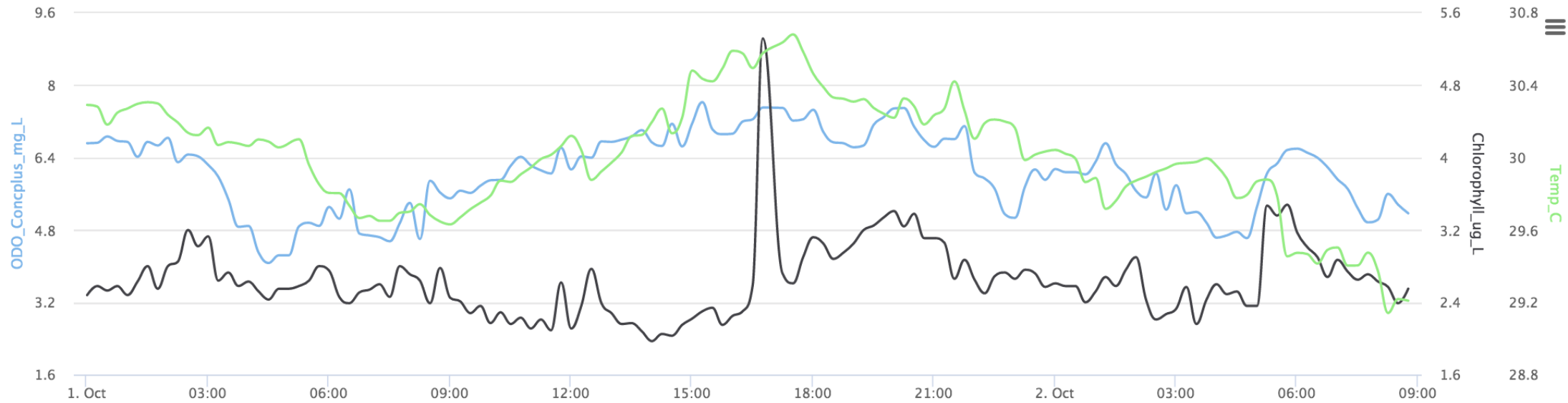
Y-axis scaling

Min

Max

Clear

✕



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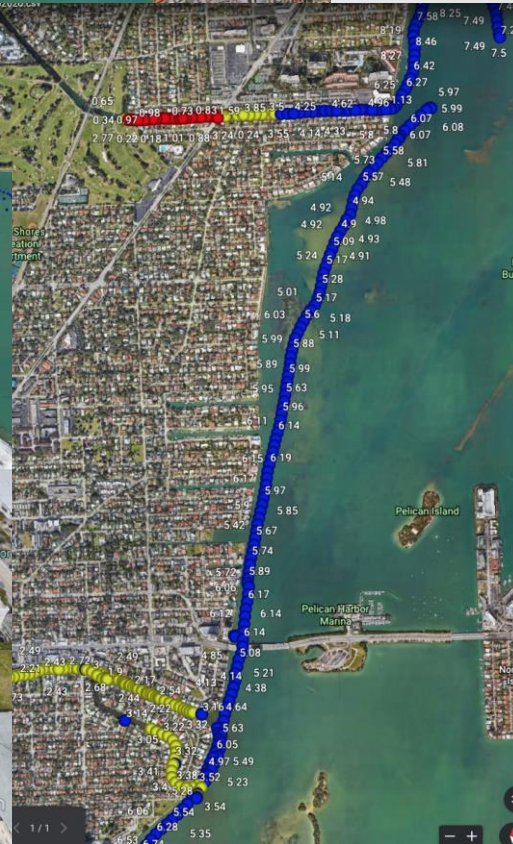
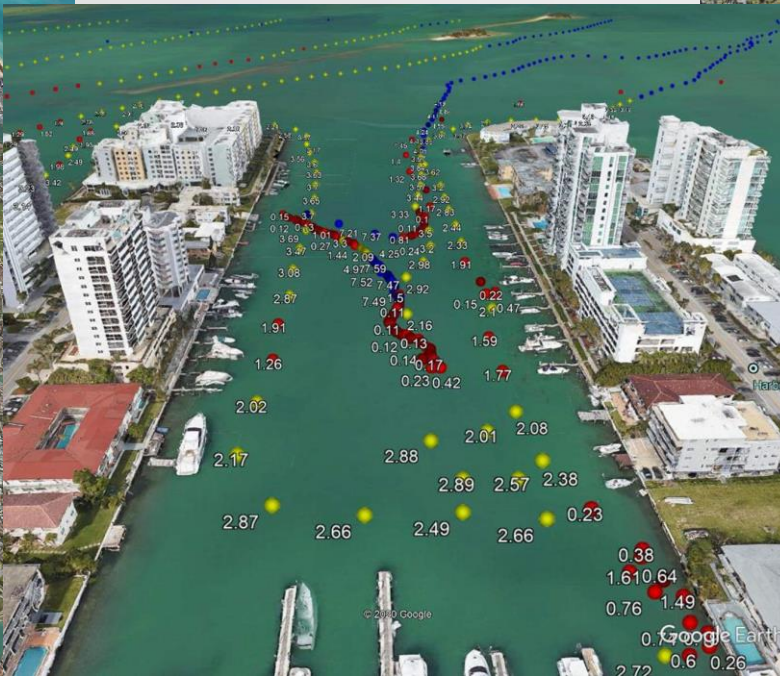
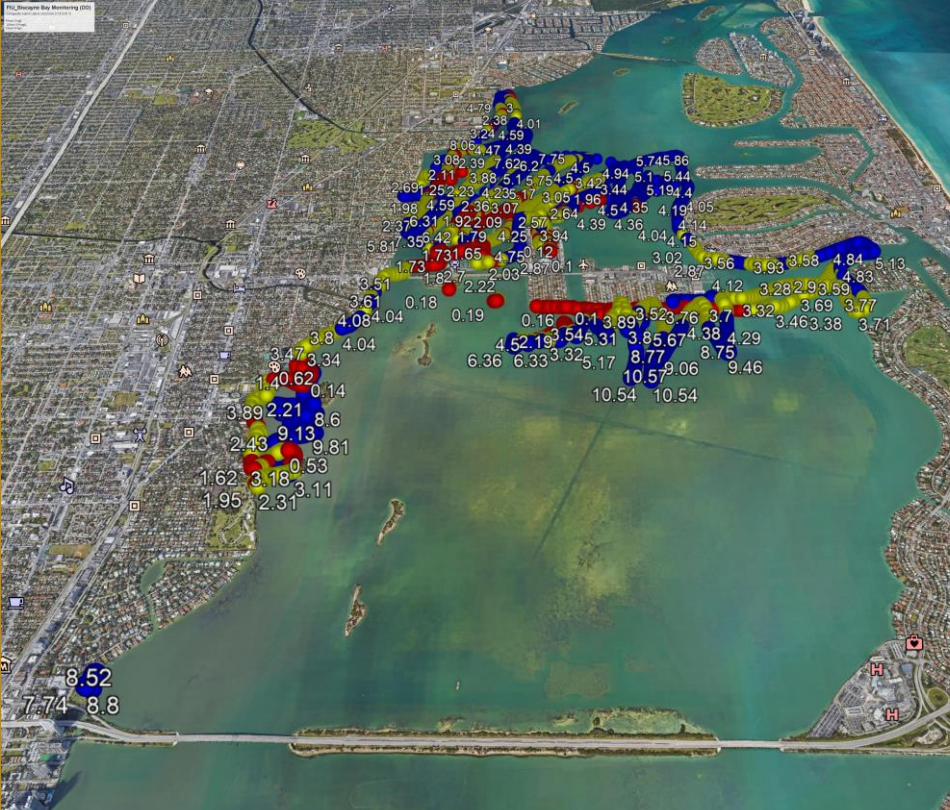
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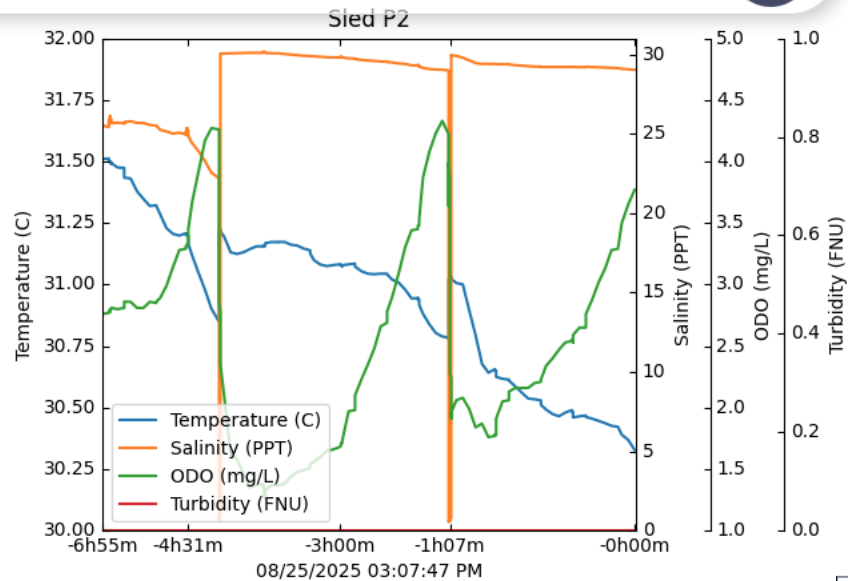
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Command



see more

Download data

from: 1 days ago

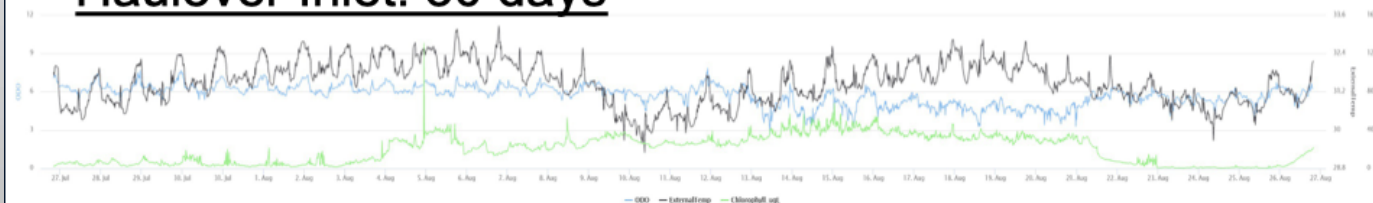
<https://crestcache.fiu.edu/research/research-buoys>

Oxygen Temperature Chlorophyll-a

Little River: 7 days



Haulover Inlet: 30 days



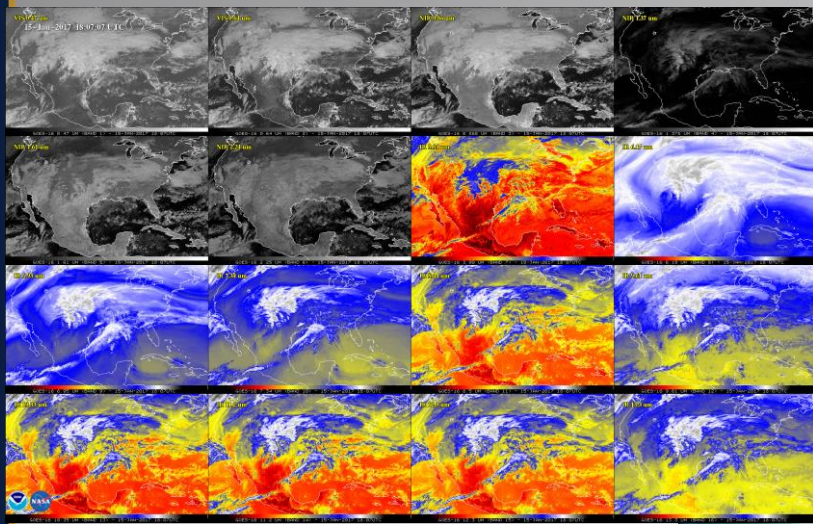
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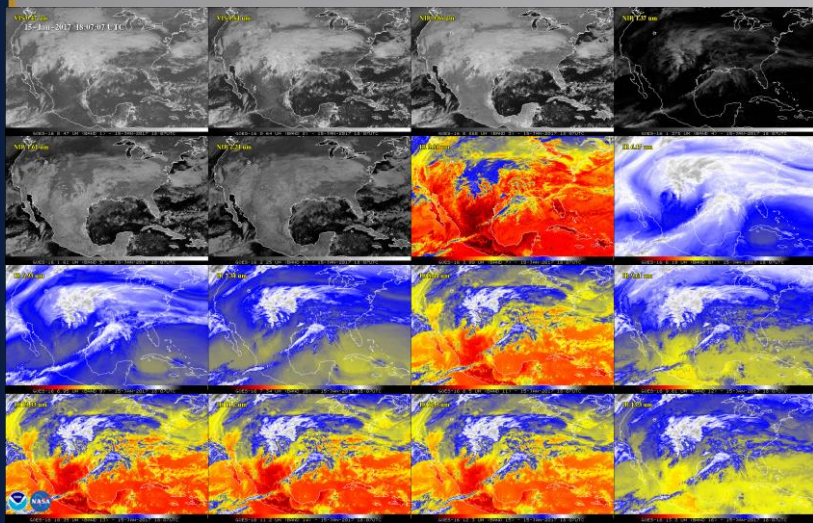
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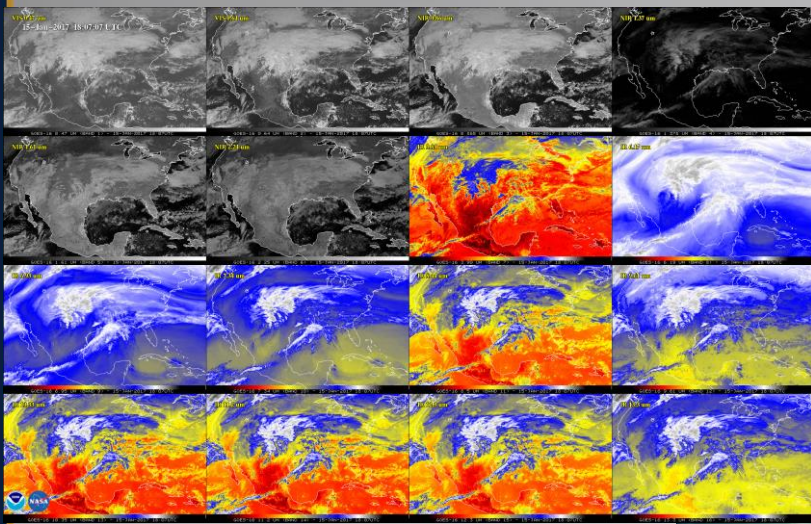
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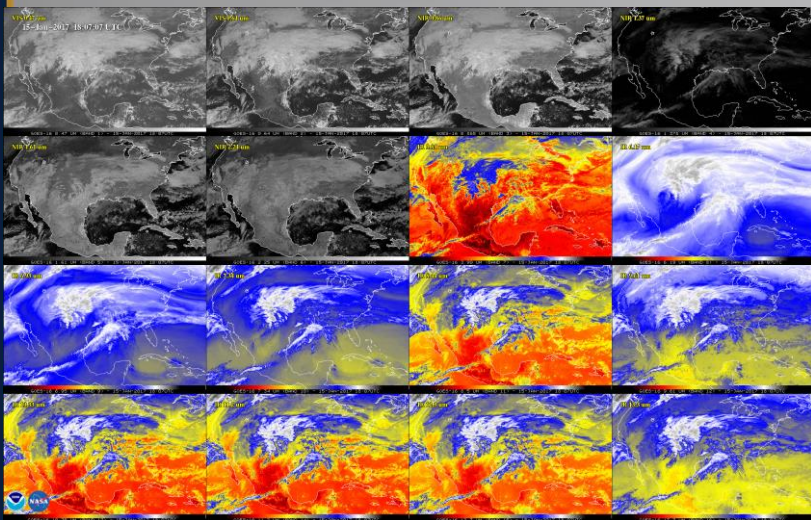
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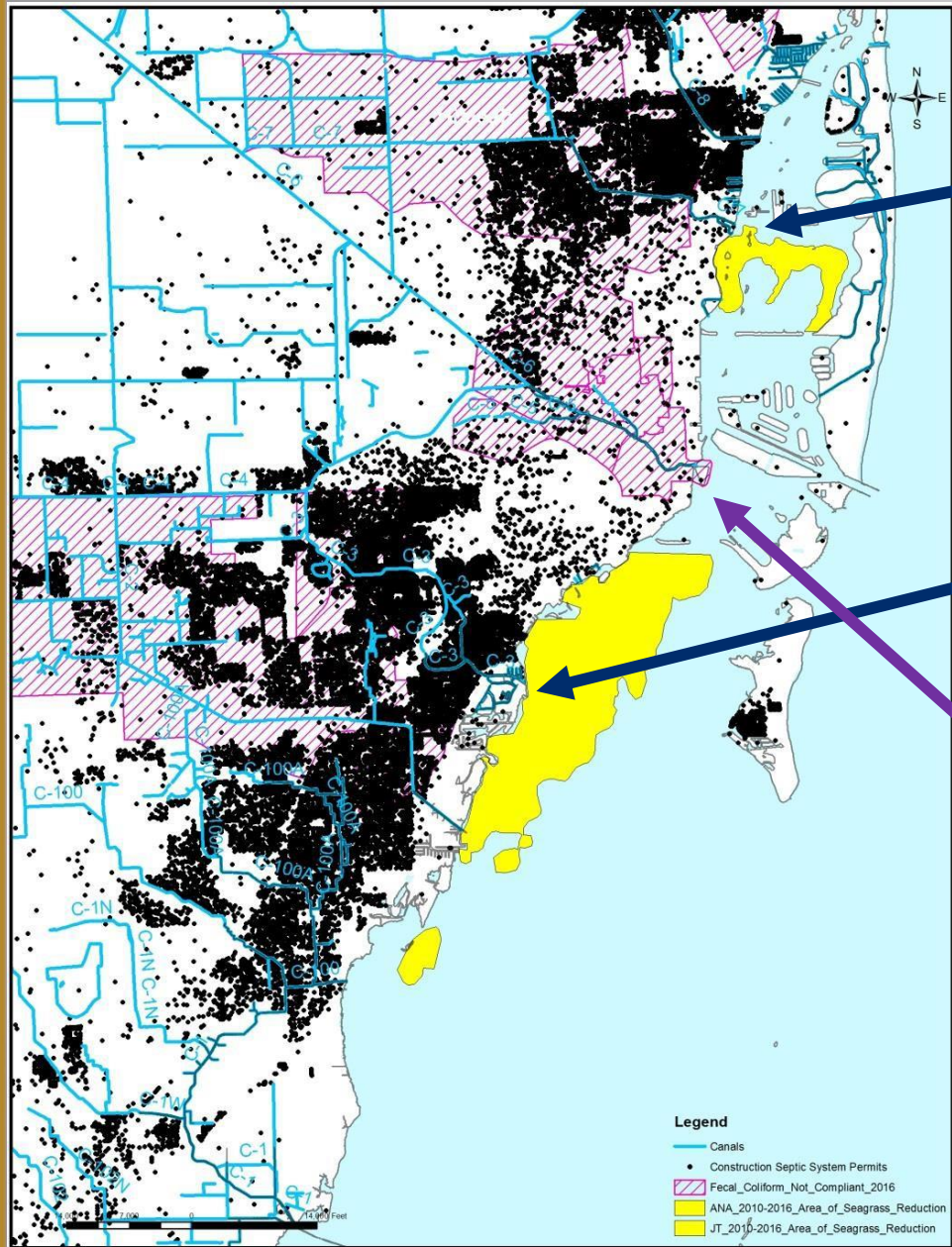








Water Bodies Impaired for Fecal Coliform and Septic Systems



- **Little River /C-7 Water** - 20% of Fecal Coliform samples exceeded 400cfu in the Body for 2016.
- **Coral Gables Waterway / C-3 and Snapper Creek / C-2** also impaired with 20.4% and 15.6% of the F. coli samples exceeding the standard.
- **Miami River Watershed**

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Biscayne Canal

Little River

Miami River

Coral Gables

Haulover Inlet

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We need a makeover:

Infrastructure

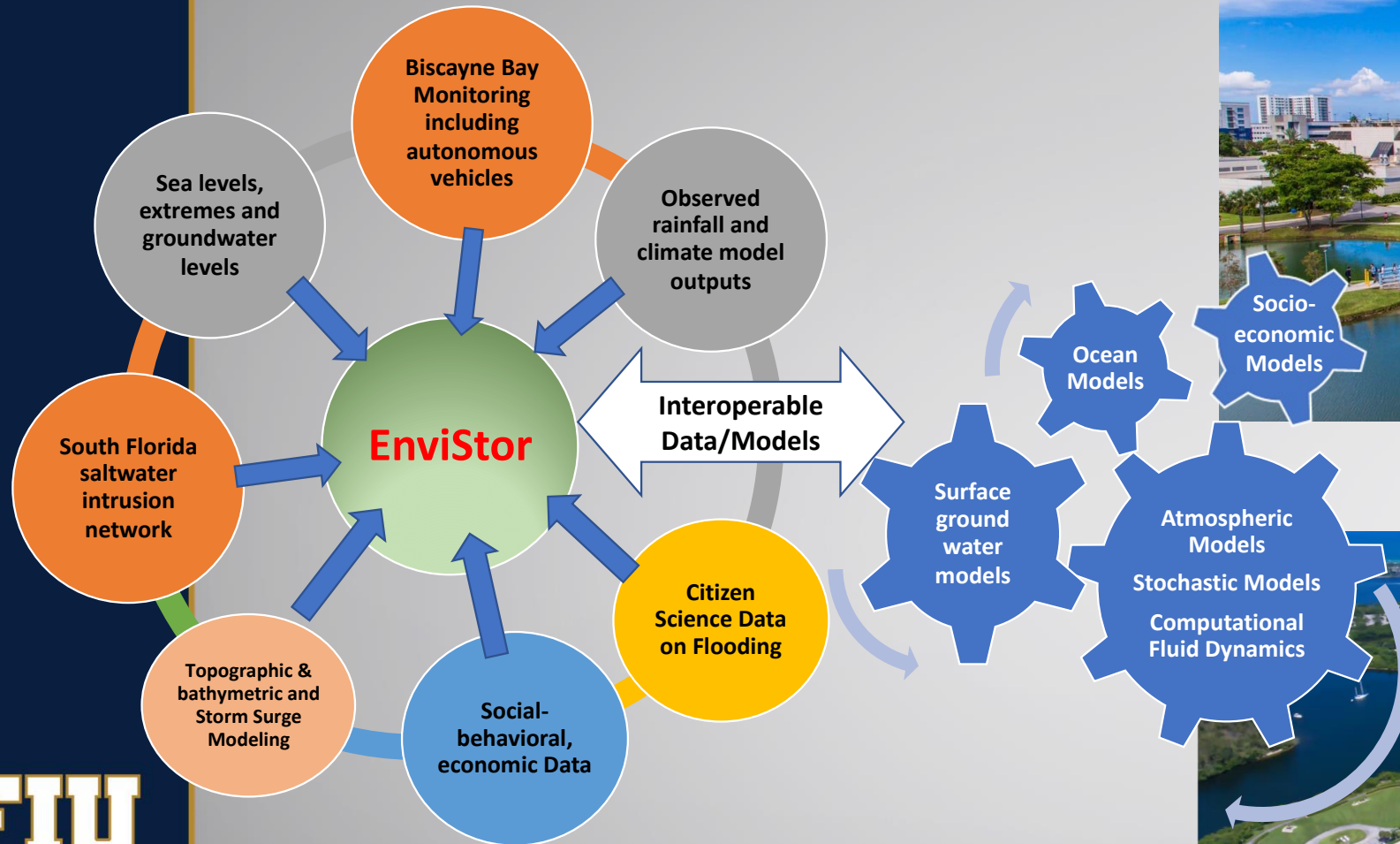
septic to sewer conversion (social justice issue)
impermeable substrates to permeable

Green Infrastructure

mangroves
sea grass restoration
native plants inland



Coastal Environmental Data Center



FIU Modesto A. Maidique Campus

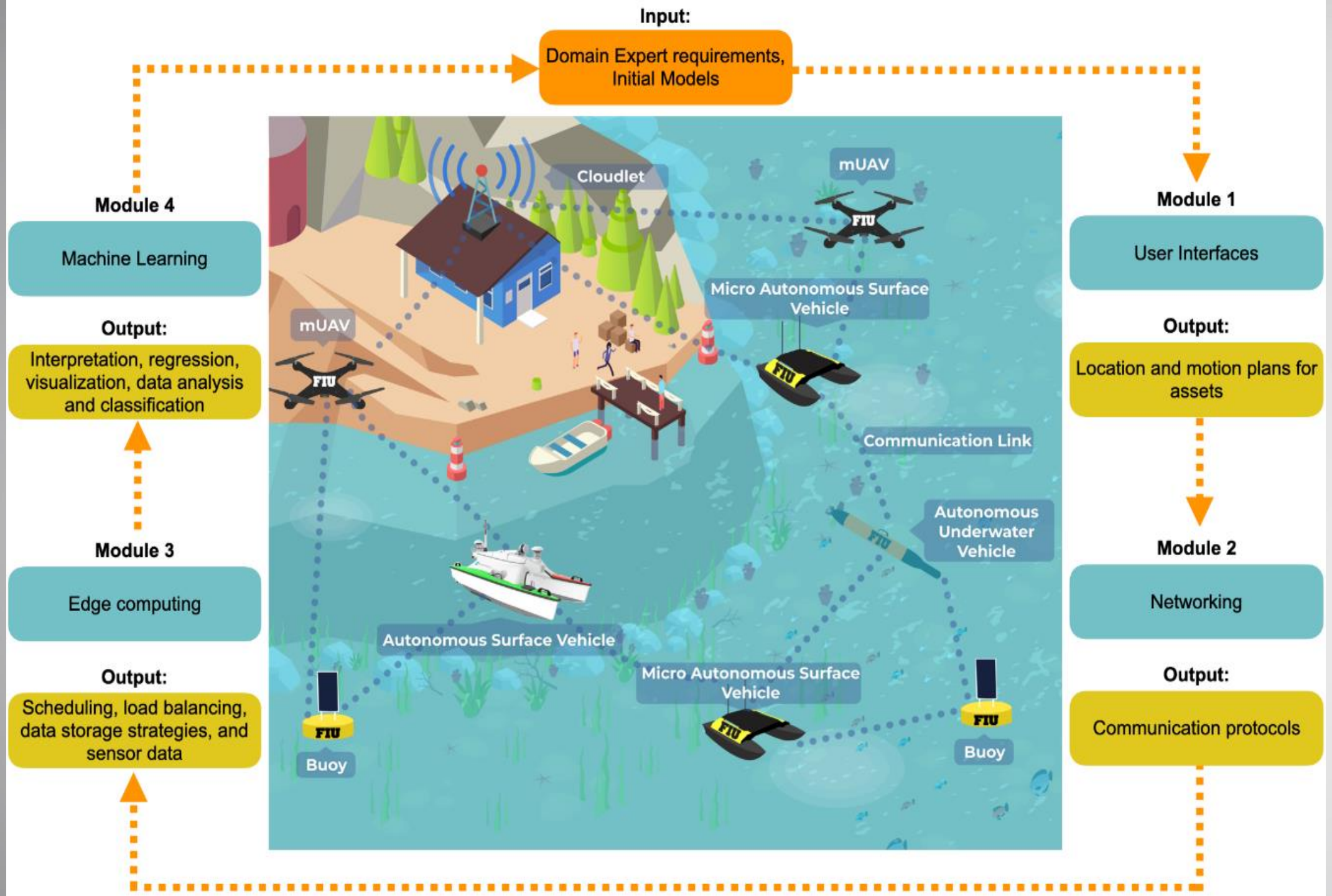


FIU Biscayne Bay Campus

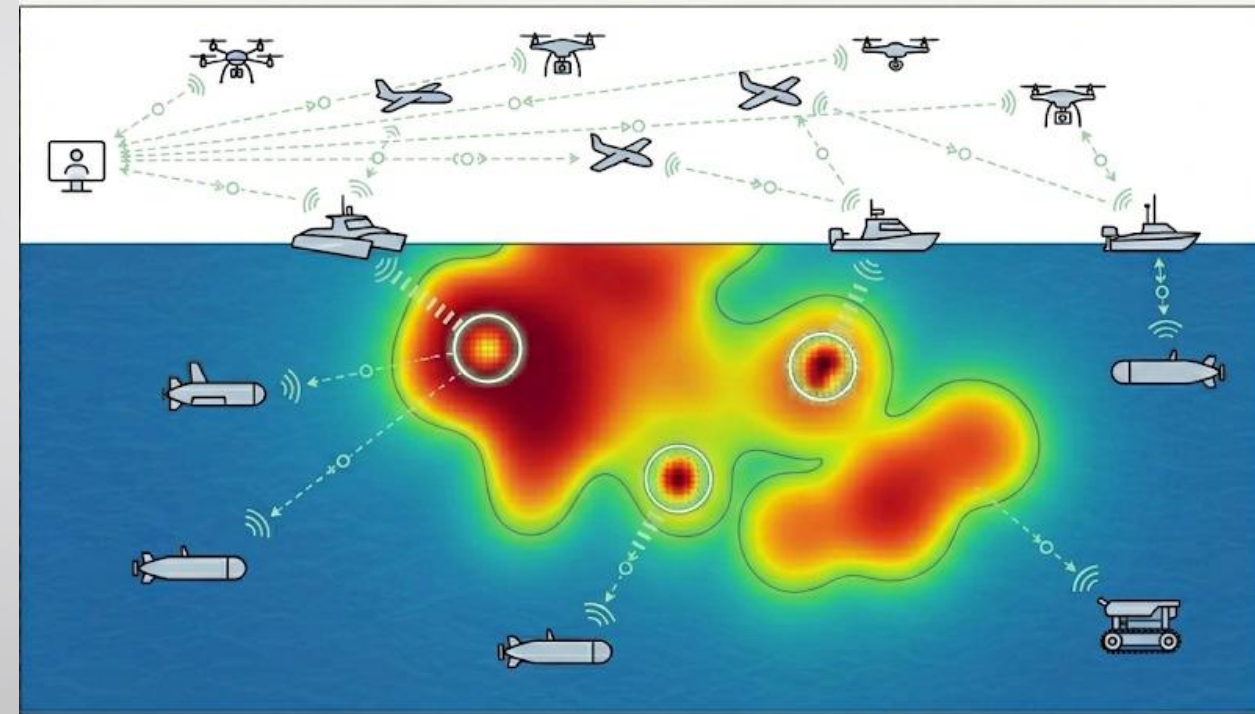
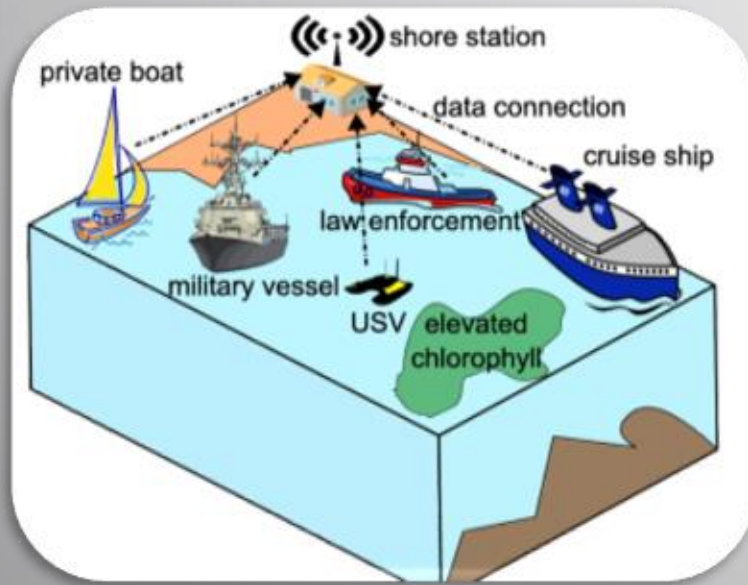
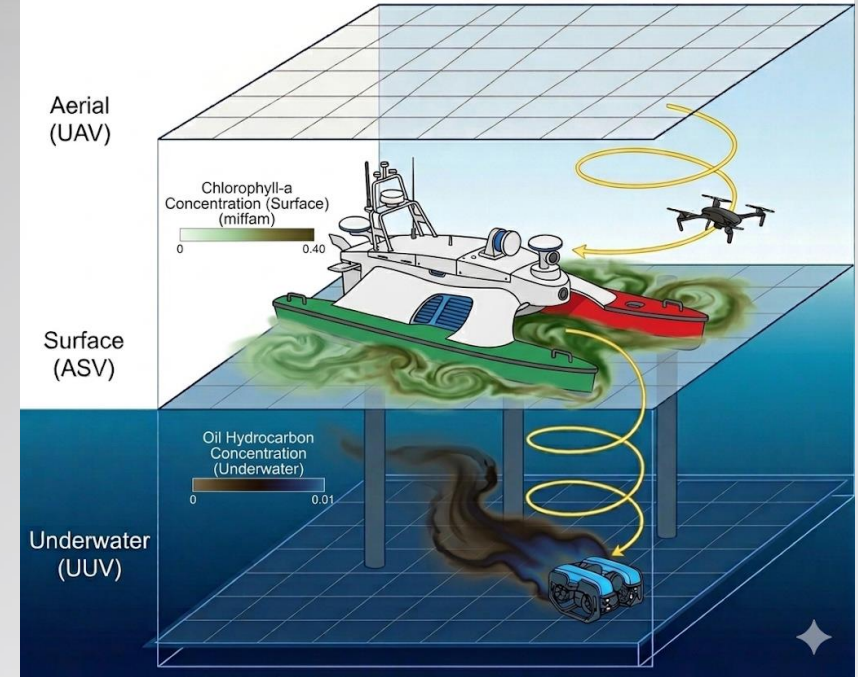


National Institute of Standards and Technology Award - \$9.2 Million

**Modular design, could be expanded
Understory parking/space
Permeable pavement
Test/Buoyancy tank on ground level**



- ★ **Critical infrastructure failures are high-impact and time-sensitive.** Where delayed detection risks safety, cost, and environmental damage.
- ★ **Current approaches rely on static or periodic sensing,** failing to adapt and missing localized, time-critical events.
- ★ **Event-driven multi-robot systems enable adaptive sensing,** reallocating sensing resources to detected regions of interest, improving efficiency and responsiveness



Using AI to guide the exploration

- Wealth or long-term data with many parameters –largely unexplored
- Novel technologies = high frequency, real-time, dense datasets
- We need to close critical data gaps
- We need to make timely decisions
- We need to think outside of the box
- **AI is just another “tool”**

We have criteria, could we explore where we are “quickly”

Today MDLs 0.0007-0.002 mg/L

Florida Administrative Code. Rule 62-302.532

Estuary-Specific Numeric Interpretations of the Narrative Nutrient Criterion

	Total Phosphorous	Total Nitrogen	Chlorophyll a
(h) Biscayne Bay	Criteria expressed as annual geometric means (AGM) are not to be exceeded more than once in a three year period.		
1. Card Sound	0.008 mg/L as AGM	0.33 mg/L as AGM	0.5 µg/L as AGM
2. Manatee Bay – Barnes Sound	0.007 mg/L as AGM	0.58 mg/L as AGM	0.4 µg/L as AGM
3. North Central Inshore	0.007 mg/L as AGM	0.31 mg/L as AGM	0.5 µg/L as AGM
4. North Central Outer- Bay	0.008 mg/L as AGM	0.28 mg/L as AGM	0.7 µg/L as AGM
5. Northern North Bay	0.012 mg/L as AGM	0.30 mg/L as AGM	1.7 µg/L as AGM
6. South Central Inshore	0.007 mg/L as AGM	0.48 mg/L as AGM	0.4 µg/L as AGM
7. South Central Mid- Bay	0.007 mg/L	0.35 mg/L as AGM	0.2 µg/L as AGM
8. South Central Outer- Bay	0.006 mg/L as AGM 	0.24 mg/L as AGM	0.2 µg/L as AGM
9. Southern North Bay	0.010 mg/L as AGM	0.29 mg/L as AGM	1.1 µg/L as AGM

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Using AI to guide the exploration

- BB is “phosphorous limited”
- Used a 20 year dataset from MDC-DERM
- AI tools will gather additional info as needed
- Weather patterns, tides, rainfall
- Promoted for few basic questions:
- Consistent “hot spots”
- Land-based?
- Stats, geographic extent, other drivers

Here are some answers and take home lessons

Highest Phosphorus Spikes in Urban Canals

The most extreme Total Phosphorous (TP) levels occurred in urban canal sites (e.g. Snapper Creek), reaching approximately 0.4–0.5 mg/L, far exceeding typical background levels. These spikes were episodic and correlated with storm runoff and wastewater indicators.

Most Samples Show Low TP Concentrations

The majority of TP measurements were very low, often near or below the detection limit (~ 0.01 mg/L). Median TP across all sites is only ~ 0.02 mg/L, highlighting an overall oligotrophic system with infrequent high-nutrient events.

Rainfall & Runoff Drive Nutrient Surges

High TP events coincided with heavy rainfall and runoff, indicating that stormwater carries nutrient loads into the bay. These events typically occur in the wet season, causing short-term TP peaks followed by a return to baseline low concentrations.

- Prompting is quite important –practice!
- Read answer carefully –be critical
- Best suited for modeling and statistics
- Good results for data curation
- As always “garbage in garbage out”
- AI “will” surprise you – alternative finds

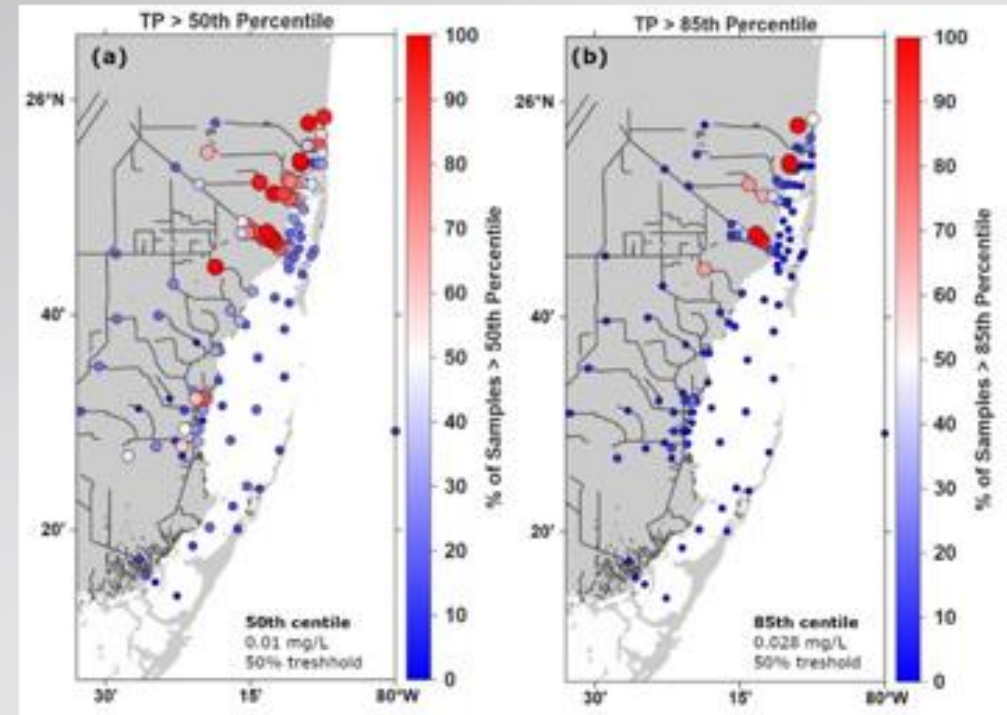


Figure 1. Percentage of samples where Total Phosphorus (TP) concentrations exceeded (a) the 50th percentile (0.01 mg/L) and (b) the 85th percentile (0.028 mg/L). In both panels, white-to-red shading highlights stations where exceedances occurred in more than 50% of the observations. Data source: DERM.

We thank you for your time



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